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Bachelor Thesis

Towards Self-Organization among Energy Cells
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Declaration of independence

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I hereby declare that I have written the present thesis independently, without assistance from external parties and without use of other resources than those indicated. The ideas taken directly or indirectly from external sources (including electronic sources) are duly acknowledged in the text. The material, either in full or in part, has not been previously submitted for grading at this or any other academic institution.

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Abstract

The widespread deployment of decentralised renewable energy sources (DRES) and the concurrent rise in the demand for (sustainable) energy are pushing the traditional centralised electricity grid to its limits. The resulting complexities of long transmission routes, many feeders and consumers and their centralised control are overloading the grid and increasingly lead to load disconnections and feed-in limitations.

Self-organised energy cells offer a promising solution to these challenges. They restructure the grid in a bottom-up approach and implement distributed control between the grid areas. The energy system, therefore, consists of self-similar cells that provide essential grid services and facilitate energy exchange via decentralised autonomous trading. In the course of the thesis, a theoretical concept for energy cells initially presented by the German Association for Electrical Engineering, Electronics and Information Technology (VDE) is expanded to include principles of self-organisation in order to cover the cell's autonomy. Furthermore, the thesis proposes a decentralised energy trading strategy derived from this cellular structure, which uniquely accounts for grid losses, thereby prioritising regional energy exchanges. This strategy is essential for mitigating grid strain and circumventing the need for extensive grid expansions. The main goal of this approach is to balance the total available energy supply as regionally as possible by not only considering the costs of energy generation but also the respective transportation costs to the consumer. Based on the developed concepts, a simulation model is implemented using agent-based modelling and simulation (ABMS). It evaluates the energy trading algorithm on two types of different power grid topologies from realistic data sets with convergence and required regionality as its main investigation criteria. In addition, the structural influences of the grid type on these criteria are analysed by comparing the observations on the extra-high voltage grid with a radial semi-open medium-voltage grid. The results of these simulations show that the novel energy trading algorithm converges and pricing grid losses secure regional energy consumption. Moreover, it is discovered that a simple random distribution of this pricing already leads to a power supply coverage of approximately 50% in a meshed energy grid. The evaluation also points out that the grid type can significantly influence the speed of convergence and the scope of regional energy trading. In Summary, this work demonstrates how a new approach to regional energy trading in the electricity grid could be realised by extending existing concepts, such as the cellular energy system and principles of self-organisation with a distributed energy balancing algorithm that considers grid losses.

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1. Motivation

The electricity demand has risen steadily in recent years. In order to meet this demand and comply with the provisions of the Paris Climate Agreement, an increasing proportion of energy is being produced decentrally by small, volatile generation plants, so-called decentralised renewable energy sources (DRES). Photovoltaics and wind energy are well-known representatives of these. However, these new types of feed-in sources also pose several challenges. The decentralised supply and the resulting increase in control commands pose considerable problems for the existing centralised electricity grid. The longer transport routes and general grid overloading resulting from distributed feed-in are further consequences fuelling calls for large-scale grid expansion. However, as large sections of the population do not desire this, and it is also very cost-intensive, alternative approaches must be considered.

One of the most promising approaches is decentralising the electricity grid into (autonomous) sub-grids. Researchers and politicians have increasingly focussed on this topic in recent years and have brought together numerous concepts under the umbrella term SmartGrids. For example, the European Union's Smart Grid Mandate (M/490) is intended to moderate this change. However, while these approaches are often only discussed in theory, there is a lack of research into the real implementation of a solidarity-based interconnected grid with decentralised, autonomous, and, above all, regional energy trading. This is needed to prevent grid overloads and the associated grid expansion.

This paper, therefore, raises the question of how such a concept could be realised. In other words, how can such regional energy trading be implemented while considering grid losses? Subsequently, the influence of different grid structures on this trading approach is also tested.

In order to achieve the autonomy of the sub-grids, an existing theoretical approach to the development of a cellular energy system is extended to include the principles of self-organisation. The focus is only on the basic concepts of self-organisation, such as the implementation of autonomy through local information and interactions. Other auxiliary services, such as self-maintenance or self-healing, are excluded for the time being. The next step will focus on autonomous energy trading. Here, transport costs in the form of grid losses or line resistance are included in the production price. The resulting dynamic grid fee is intended to price the transport costs and thus favour regional energy equalisation. The novel approach developed is then examined regarding convergence and regionality using agent-based modelling and simulation. The work aims to demonstrate a possible implementation of the energy trading described above and provide a practical simulation of the approach. For

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now, grid services such as frequency maintenance or outage management are not considered.

Section two discusses essential background information. Firstly, the basic existing concept of the cellular energy system is presented. Secondly, the basic concepts of self-organisation are discussed, which should then add autonomy to the cells of the energy system. Section three then explains the actual implementation and customisation of the cellular structure and presents the novel approach to regional autonomous energy trading in the cellular energy system. A subsequent simulation and the subsequent evaluation and discussion of the results are presented in section four. The simulation tests the developed approach on two different grid structures. The convergence and regionality of the energy trade achieved are analysed. Finally, Section six concludes and provides an outlook for further work.

2. Background

In order to understand the developed approach to self-organised regional energy trading, necessary background knowledge is given. This includes a brief overview of the current centralised electricity market, its fundamental points and physics, as well as the VDE concept of the cellular energy system and important principles of self-organisation.

2.1. Fundamentals of the electricity grid and the electricity market

The following section provides an overview of the traditional electricity market featuring core concepts such as flexibility and the merit order price. Additionally, impedance gets introduced as a parameter for grid losses.

2.1.1. Current centralized electricity market

With the liberalisation of the electricity sector in the EU, all member states have restructured their electricity sector from a generation, transport and distribution monopoly to a central electricity market with many suppliers. Transmission System Operators (TSOs) and Distribution System Operators (DSOs) were introduced as part of this process. They are regulated and responsible for the operation of the grids and associated grid services.

Independent national single markets that import and export electricity to and from each other form the European energy market [Pol19]. Within the countries, there are in turn a large number of different markets. There are "forward markets", "day-ahead markets", "intraday markets" and also "continuous day-after markets". Energy is traded centrally in these markets at different times to cover forecasts or, in some cases, actual consumption. Grid operators (DSOs) are responsible for preparing these forecasts. However, as the forecasts of future consumption are less than one hundred per cent accurate, it is also their task to moderate possible grid bottlenecks. The grid operators must ensure that the grid frequency does not exceed or fall below 50 Hz at any time; otherwise, the grid will collapse. In contrast to the market, they, therefore, take account of the actual balance between supply and demand in their grid through flexible loads and generation. This balancing always

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takes place at quarter-hourly intervals. A central control centre controls the switching on or off of loads or reserve power plants.

This means that the grids currently have centralised control and also centralised energy trading on the respective markets. Furthermore, no grid losses are currently included in the generation pricing on the electricity markets. This means the currently implemented approach does not consider regional generation or consumption. For example, the energy from large wind farms in the north of Germany flows via power lines to the best supply in the country's south. As electricity consumption increases, an increased need for socially undesirable grid expansion is needed.

2.1.2. Flexibility

The more distributed renewable energy sources (DRES) push conventional fossil-fuelled power plants out of the energy market, the more critical flexibility in the associated grids becomes. In addition to cheap energy, renewable energies bring two key challenges.

The first is variability. The generation of DRES is often very dependent on external factors such as wind or sun. For example, the weather determines the current maximum generation of DRES and changes this over time. The other is uncertainty. Despite good weather forecasts, it is impossible to make a perfect prediction about the future generation volume of DRES. The energy grid must, therefore, be prepared for these two challenges. Flexibility is an essential tool for that.

The Federal Network Agency of Germany (BNetzA) defines flexibility as follows:

“On an individual level flexibility is the modification of generation injection and/or consumption patterns in reaction to an external signal (price signal or activation) in order to provide a service within the energy system. The parameters used to characterise flexibility include the amount of power modulation, the duration, the rate of change, the response time [...]” [\[bne17\]](#)

Flexibility, therefore, means that a technical unit can reliably provide changes in its output, either in the form of a change in feed-in or in the form of a change in energy consumption.

The unit must guarantee the following points:

- **Magnitude:** It must be able to reliably and constantly draw or feed in the required amount of energy.
- **Ramp rate:** The required feed-in or offtake must begin after a defined start-up time. This start-up time must be very short to ensure a high degree of flexibility. Conventional fossil-fuelled power plants, on the other hand, are characterised by a very long start-up time and, therefore, by less flexibility.
- **Duration:** The feed-in or offtake must occur over a certain period of time. In other words, a certain amount of energy per time must be guaranteed.

Flexible generation or consumption is essential to ensuring a stable energy grid. It plays a particularly important role in the sustainable decentralised electricity grid. Different technologies can provide this required flexibility. On the one hand, traditional energy storage systems such as batteries. On the other hand, sector coupling can also provide this. Power-2-X technologies convert different energy sources into each other, making energy storable and usable at a later date. However, flexible generators or loads, such as DRES, also contribute to the flexibility of an electricity grid. Rebound is another general form of flexibility. Here, an amount of energy is generated or consumed at a selected time so that it no longer needs to be considered at inappropriate times. For example, if an electric car is charged at night, charging in the morning is no longer necessary.

2.1.3. Impedance

The electrical impedance or complex resistance is the equivalent of the ohmic resistance of the direct current in alternating current technology. Similar to Ohm's law $R = \frac{U}{I}$, there is a law that relates this resistance ($\underline{Z}$) to the AC voltage ($\underline{u}$) and the AC current ($\underline{i}$) [Ken93]¹.

$$\underline{Z} = \frac{\underline{u}}{\underline{i}}$$

The essential difference between ohmic resistance and impedance is that the latter considers a possible phase shift between alternating current and alternating voltage. Identical to ohmic resistance, a high impedance between a current source and a sink produces a high energy loss between them. This means that the impedance can also be seen as an electrical distance. A low electrical impedance corresponds to a small electrical distance, while a high impedance corresponds to a greater one.

Distance should not be regarded as a geographical unit of measurement. It is more of a parameter for the additional energy required to transport the charges from point A to point B. An excellent example of this distinction is that a geographically longer power line with a thicker diameter can have a lower impedance than a geographically shorter line with a much thinner diameter.

In the electricity grid, the impedance of the individual lines is a decisive factor in the level of grid losses. If a line has a high impedance, a current flow via this line automatically results in a high energy loss. However, it should always be noted that electricity always takes the path of currently least resistance, i.e. it is divided between several power lines and does not just choose the line with the lowest impedance. This means that to measure the exact resistance between two points, a power flow analysis of all lines with the associated impedances would have to be carried out.

¹Formula symbols changed to standardized symbols

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2.1.4. Merit-Order Principle

The merit order principle is a model of the central electricity market. It is used to calculate the current market price for electricity. At fixed times (usually every quarter of an hour in the day trading market), the available electricity generators are ordered according to their marginal production costs. As can be seen in Figure 2.1, the production blocks are ranked in ascending order up to the block with the highest production costs. The grid operators forecast an electricity demand in advance. The merit order principle now only accepts blocks before or at this capacity limit. Only the most favourable offers are accepted to cover the demand. The price of the last accepted offer determines the current global electricity price. This is also called the clearing price, as it clears the market. This means that although, for example, in Figure 2.1, DRES produce much more cheaply, their electricity costs the same as the last accepted gas block. The electricity producer can book the difference as a profit. The merit order principle was designed to remove the most expensive electricity producers from the market, ensuring cheap electricity prices free from speculation. This model also benefits distributed renewable energy sources (DRES) by encouraging their use through high profits. This, in turn, lowers the clearing price for all consumers, leading to lower power prices.

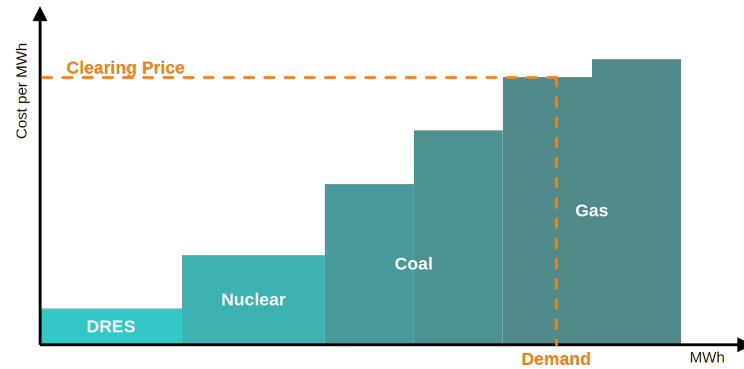


Figure 2.1.: Merit-Order principle of the centralized power market

2.2. Cellular Energy Systems

The energy landscape is undergoing profound change. Traditionally, a few large energy producers have served many consumers. However, now, they are supplied by a more significant number of smaller energy producers instead of a few large ones, such as households with PV systems.

This change from a centralized energy grid in the past, which tended to feed energy unidirectionally from higher voltage or pressure levels to lower levels according to the top-down approach, to a decentralized grid in which energy is mainly fed in and consumed at the lower distribution grid level, must be designed accordingly. The

decentralized balance of generation and consumption at the lower levels is becoming increasingly important. It implicitly leads to subdividing parts of the energy grid into so-called cells. The concept of self-organized energy cells intends to optimize grid utilization, minimize grid overloads and facilitate the integration of renewable energy sources. In addition, the decentralized nature of energy cells enables efficient energy use at all levels - local, regional and national.

A well-known approach for the distributed cell system is the MicroGrid. It describes that parts of the grid can operate autonomously and independently of the primary grid due to their generation and consumption. However, while this approach only refers to the electricity grid and focuses on the individual cell rather than the network of cells, the Association for Electrical Engineering, Electronics and Information Technology (VDE) approach refers to a more holistic concept that considers the cellular energy system as a grid divided into various co-operating self-organized energy cells and defines the individual cell as follows:

“An energy cell consists of the infrastructure for different forms of energy in which energy cell management, in possible coordination with neighboring cells, organizes the balancing of generation and consumption across all available forms of energy.” [JB19]

Although the cells in the energy grid vary in size, from very small cells, such as a household, to very large cells, such as an industrial park or a large city, they all have a self-similar structure and are, therefore, defined identically [TB15]. The definition places several requirements on each cell, briefly explained below.

Infrastructure: A cell’s infrastructure includes all resources and processes for generating, transporting, distributing, and storing energy within the cell. A cell can contain all, several, or just one form of energy, such as electricity, gas, heat, or mobility.

Sector coupling: Sector coupling describes all technologies and processes for exchanging the various forms of energy, such as electricity, heat, gas and mobility, with each other, including all Power-2-X technologies. If a cell has several forms of energy, it also balances its demand and supply across the different forms of energy.

Principle of Subsidiarity: One of the two most important properties of the energy cell is subsidiarity. It requires that any control deviations or problems are primarily dealt with directly at the source and secondarily in the nearest neighboring, upstream, or downstream grid areas or cells. This includes ensuring that the physical balance between energy supply and demand is established as far as possible at the regional or local levels. This will accelerate the expansion of renewable energies and significantly relieve the transmission grid.

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Principle of Solidarity: The other of the two most essential points is solidarity. A cell must help its neighboring cells or cells above or below to balance energy levels. At the same time, however, it can also plan with their help to compensate for its energy deviations, so unlike in some microgrid approaches, a cell must never isolate itself in regular operation but must constantly interact and trade with its neighbors.

States: Each cell always has one of the three states: balanced, oversupplied, or undersupplied, which it calculates using all forms of energy. Furthermore, each cell is categorized hierarchically according to its size and role in the energy grid into one of five levels—local, regional, supra-regional, national, or international, as seen in Figure 2.2. Data and information processing always takes place at the lowest level, per the subsidiarity principle, to prevent capacity problems or time delays caused by shifting too much work to the upper levels.

Goal: Regardless of its status in the system, each cell has an identical goal: to autonomously balance the generation and consumption of energy within the cell or with the help of other cells using flexibility options. It is, therefore, not a question of striving for maximum self-sufficiency but of exchanging generated power and flexibility options to cover the entire load and avoid having to switch off generators.

Energy cell management (ECM): A cell-internal management system maintains all these aspects. It coordinates and controls all components, such as DRES, and communicates with other cells. The ECM of each cell plays the role of the brain, as this is where all data and system services converge. Data gets aggregated, and predictions are made based on this real-time data, under which storage or sector coupling technologies can be used or purchased in a local or regional flexibility market. The cells communicate exclusively via the ECM and only use the aggregated data to guarantee security and efficiency. The ECM can also prescribe frequency regulation, load reductions, feed-in reductions or other grid measures to avoid curtailment for subordinate components or cells.

Figure 2.2 shows that a large number of the cells just defined make up the cellular energy system. They are repeated at all grid levels and span the five previously described cell levels from low voltage or pressure, i.e. the local level, to maximum voltage or pressure, i.e., the international level. The system, therefore, builds up from the bottom of individual cells. These are then enclosed or clustered by cells in the layers above. The system is, therefore, a nesting of cells. Nevertheless, all cells must be self-similar, i.e. implement and realize the abovementioned aspects. The main principles of solidarity and subsidiarity lead to higher-level cells having a precise grid status and flexibility information about their subcells.

The electricity grid will form the basis for the cell division, as electricity cells will be defined first based on the physical and regional conditions. These will then be

expanded to include gas and heat to form energy cells. Characteristics for the cell type are the structure type (industry or settlement), maximum output (load and feed-in), annual energy, storage (output and capacity), grid data, flexible load, and sector coupling.

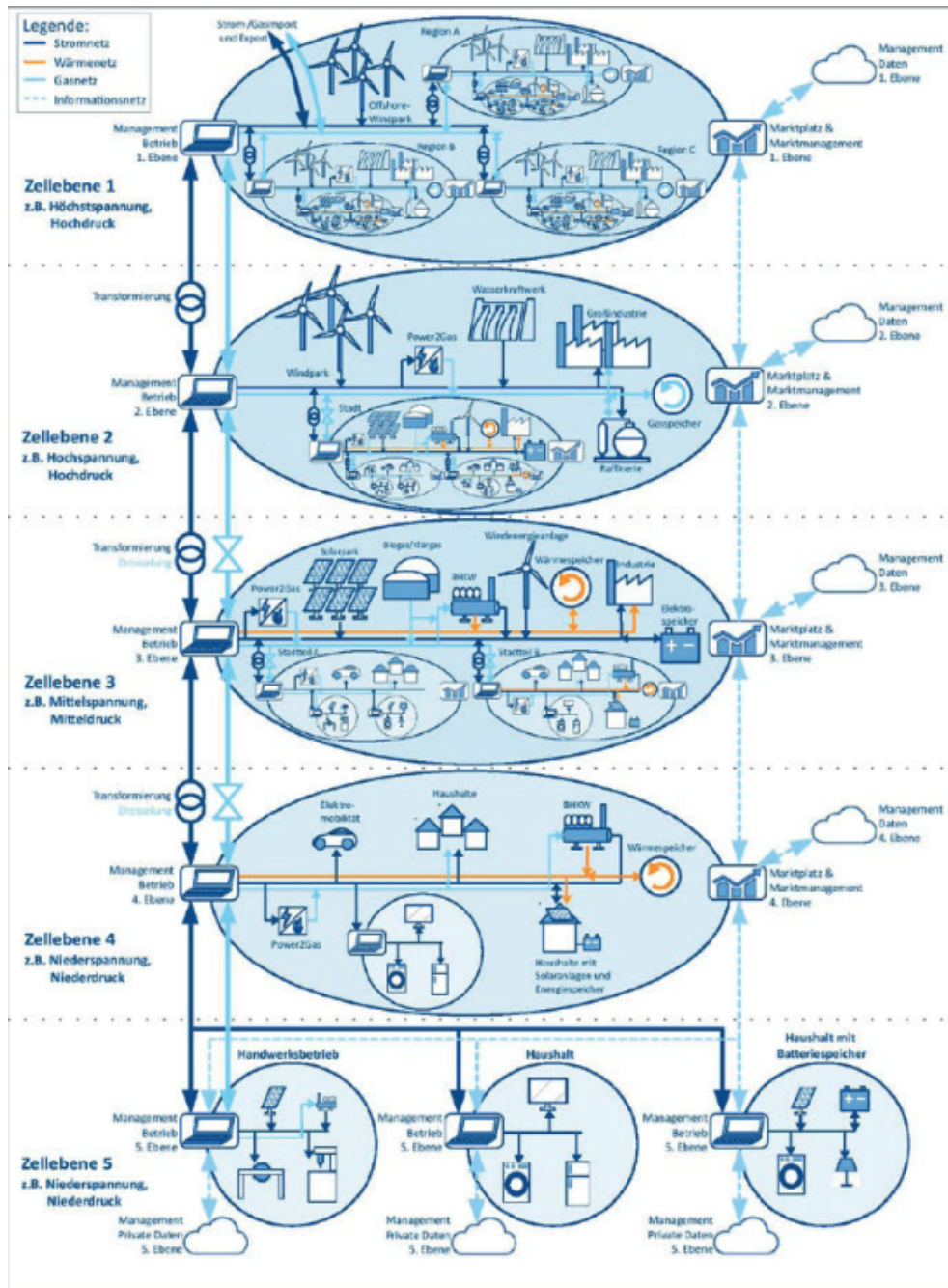


Figure 2.2.: System architecture of the cellular energy grid developed by the VDE

2.3. Self-Organization

Self-organisation plays an essential role in the energy grid of the future. Due to the constantly growing number of controllable DRES and the increased amount of data to be transmitted, the original centralised system is becoming increasingly inefficient. Therefore, the research must search for a new approach that is more scalable and avoids long transmission and response times. Self-organising systems are a promising approach for use in the energy grid, as they have the required properties such as being scale-free [DMK05], robustness and resilience [DMSFH⁺04]. In other words, “[...] large-scale complex systems [such as the energy grid] have a high desirability to be self-organising [...]” [YZV17].

“Spontaneous emergence of global coherence out of local interactions”
[H⁺01]

This is how Francis Heylighen et al. define self-organization in his much-cited seminal work. However, the term originated much earlier. In 1962, Ashby introduced the term “self-organization”, meaning some pattern is formed by the cooperative behaviour of individual entities without any external control or influence [Ash17]. Since this definition, self-organization or aspects of it have permeated almost every subject area, from describing natural processes, such as the well-known example of describing an ant’s nest [DMSFH⁺04], to technology [H⁺01] and software development [YZV17]. It is, therefore, difficult to find a precise and concise definition of the term self-organization. In addition to the definitions given at the beginning, several other fundamental concepts and principles have emerged [DMK05, HDMB08].

From these different approaches, however, a core of fundamental concepts and principles has emerged that underlies all self-organizing systems. This paper, therefore, first introduces these basic concepts and then concentrates on the specialized case of self-organization in P2P systems. ABMS rounds off the section and offers the possibility of simulating and evaluating the self-organized system.

2.3.1. Basic Concepts and their Definitions

Self-organised systems consist of components, agents, and their interactions. Agents reside in an environment or neighbourhood and can interact with local or connected agents. Furthermore, each agent is limited by their knowledge and perspectives, so they have to team up to overcome their limitations and solve problems beyond their individual capabilities through local interaction. The concept of self-organisation does not allow for global interaction or global knowledge. This means that agents may only interact with a limited number of agents. These local interactions automatically lead to emergent structure and functionality.

In detail, agents perceive their environment or neighbourhood through their sensors and then perform actions through their effectors, which, in turn, affect the environment. Each agent has a goal that influences their decision for the following actions.

Self-organising systems make it possible to grow quickly and easily, as it has no central control and each agent only contributes to the global behaviour with local calculations. It is therefore scale-free.

The following points describe the requirements and properties of such a system.

Autonomy Agents must be able to operate independently of their environment. This does not mean that they have to survive without their environment in complete self-sufficiency; rather, they simply have to be able to make decisions independently of their environment. Furthermore, like any other system, every agent must be able to draw boundaries to the environment. The crucial point here is that the system itself must draw these boundaries to maintain its autonomy and not be subordinate to any external decision-making power [DMK05].

Emergence The Greeks already researched the meaning of emergence. “The whole is greater than the sum of its parts” was coined by Aristotle. Back then, emergence was considered a black box, but today, the research sees it as a white box (neo-emergentism) [SIK06], in which global order is created through local interactions. To put it better, local interactions without global knowledge or structure add up to a globally recognisable structure at a higher level. Therefore, the individual agent cannot perceive emergence; only an observer outside the system recognises it [DMSFH⁺04]. One way to understand emergence is to consider how local interactions initially affect only the immediate neighbourhood but then trigger a chain reaction that eventually manifests as global behaviour. Similar to the conservation of momentum across multiple objects in physics [H⁺01]. Well-known examples of emergence are our cell system or the game of life, in which a cellular automaton recognises a global pattern after a certain time [Ada10]. Emergence is responsible for the robustness of a self-organising system.

Autonomy and emergence are two of the most essential characteristics of self-organising systems and are also indicators of the degree of self-organisation. The level of autonomy shows how much external data is used to control the system, and the level of emergence shows how much cooperation takes place between the agents. Both levels cannot be low in a self-organising system. Richard Holzer et al. [HDMB08] have found a formal definition for these critical parameters using the notion of entropy known from information theory.

Stigmergy Stigmergy was first observed interacting within an ant colony [Gra59]. It means that agents communicate indirectly via the environment. In this case, the environment serves as a communication medium. In other words, they leave a trace in the environment that stimulates the execution of the next activity, reinforcing subsequent actions and building on each other. Stigmergy leads to emergence, among other things. The trace usually consists of pheromones, which can be

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natural (ants) or digital (tags) [DMSFH⁺04].

Although many self-organising systems implement it, the absence of stigmergy is not an exclusion criterion, and interaction among agents can occur directly instead [SIK06].

Decentralized Control The control is distributed throughout the system, and all parts contribute evenly to the resulting arrangements. Therefore, different configurations or mergers of agents can also arise in different parts of the system, which then fight against each other for global emergent behaviour or balance each other out and coexist in different sub-zones. In general, there are two different types of self-organising systems, which differ according to the type of control [SIK06]:

- Strong self-organising systems: The system is decentralised, and its agents have decentralised internal control, too. Therefore, there is no centralised control in the entire system.
- Weak self-organising systems: Agents may also have an internal centralised control unit.

However, the overall system must be decentral in both approaches.

Criticality & Equilibrium Self-organising systems are always initially disordered, i.e. no emergent behaviour is yet visible. There is a point in time or a condition under which the system transitions from the initially disorganised state to an ordered or stable one, i.e. the observer sees emergence. This point is called the critical point or criticality. It is the exact point at which the chain reaction for the emergent behaviour starts and is often influenced externally, e.g. by temperature or an intent, but can also come from within the system. It so it is then called self-organised criticality.

Although there are several stable end states, it is unpredictable exactly where the system will end. Nevertheless, the final state must always be a state of equilibrium, in which the parts of the system have balanced each other out, and the system is stable, i.e., no further cascading interactions occur. Equilibrium can be interpreted as a state where different parts of the system are mutually adapted.

Feedback Feedback is the means of control of a self-organising system. There are two different types of feedback: Positive feedback: reinforces or accelerates the development or change to another state Negative feedback stabilises the system by suppressing or counteracting change. Initially, the change process starts with positive feedback until equilibrium is reached, after which negative feedback no longer leaves this state. [H⁺01]

2.3.2. Self-Organization in P2P-Networks

In addition to the generally applicable criteria mentioned in subsection 2.3.1, there is another characteristic for self-organising systems in P2P networks, namely hierarchy and heterarchy [CSO]. Both describe how the system organises the communication among the agents and both can be used together or individually.

Hierarchy is therefore defined as a rooted tree, i.e. an undirected simple graph in which a pair of nodes can only be connected by exactly one unique path. One node plays the role of the root, while all other nodes are arranged in descending order from the root.

Heterarchy is defined as a network structure in which a node can be connected to any number of neighbours, i.e. the opposite of a tree. There is no root, and therefore no order.

While hierarchy helps to partially organise the agents and assign them a distance to the root node through the unique communication path, heterarchy enables a high degree of connectivity. The network is therefore more fault-tolerant.

2.3.3. Simulating Self-Organized Systems

A popular approach to simulating Self-Organizing systems is agent-based modelling and simulation (ABMS). Charles M. Macal describes it as "a third way of doing science and could augment traditional deductive and inductive reasoning" [MN09]. Researchers can use the approach as an e-laboratory to test and evaluate complex systems and models, like self-organised systems, in more depth and on a larger scale before impacting real-life systems [NCK⁺02].

ABMS also consists of a set of agents and a framework that simulates their decisions and interactions with each other over time. The implementation of the ABMS framework is highly flexible, accommodating a wide range of scenarios. It can be as simple as live simulations, where real people act as the agents, or as complex as using general-purpose tools like spreadsheets. The most comprehensive approach, however, involves dedicated software frameworks such as Mesa [KMC20] or RepastJ [NCO⁺13]. These frameworks offer the unique advantage of efficiently implementing and testing complex models on a large scale.

Identical to self-organising systems, agents are autonomous, self-directed entities that live in an environment, are social and interactive, have local goals and represent a specific resource. During the simulation, these agents use limited decision rules and local interaction with other agents to achieve their local goals. This continues until a desired end-state, a global equilibrium, is reached. There are different ways of modelling the social interaction between agents and determining which agent is connected to whom, when and why. Cellular Automata, developed by Stanislaw Ulam, implement indirect interaction through proximity [CD98]. The network topology realises direct communication, which corresponds to a static or dynamic network of agents and can, therefore, be independent of proximity. Other approaches, such as the Soup or Euclidean space topology, also exist and can be looked up here [MN09].

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Regardless of the topology, however, the self-organised principle always applies that interaction may only be local, i.e. only with a limited number of agents.

ABMS is used both in research and the market economy [MN05] and ranges from small minimalist models with a set of idealised assumptions to capture only the most salient features to large-scale computational clusters. It is easy to start with a small model and extend it to large scales.

2.4. Related Work

Although the energy market has become increasingly complex in recent years due to deregulation and the introduction of renewable energies, and new approaches are required, little to no research has been done on the combination of self-organisation and a cellular energy system, let alone pricing in the transport costs, such as impedance, in such a system to enforce regionality. Nevertheless, some approaches either attempt to bring self-organisation into the energy market or divide the grid into cells.

The Argonne National Laboratory’s Center for Energy, Environmental, and Economic Systems Analysis (CEEESA), for example, developed a simulation tool on behalf of the U.S. Department of Energy that implements a self-organising system based on ABMS to test new market restructurings and deregulation on the American electricity market and examine their effects [NCK⁺02]. The system, called the electricity market complex adaptive system (EMCAS), implements three types of agents: generators (GCA), consumers, and the grid operator (ISO). GCAs mimic generation companies that compete in the market and try to get the best price for the electricity they generate. They, therefore, try to maximise their company value, which is in stark contrast to our approach, mainly based on solidarity among agents. As mentioned, EMCAS simulates the behaviour of GCAs and their marketing strategies. Therefore, GCAs endeavour to use different strategies to explore the physical limits of the electricity system and the market rules under which they operate. EMCAS is, therefore, based on the existing electricity grid and does not pursue a cellular approach. Another significant difference is that the model only supports energy through fossil fuels, which also shrinks the available flexibility compared to the presented approach in this thesis.

Abdalkarim Awad et al. have also dealt with self-organisation in the smart grid but concentrate less on the typical case, in which energy is traded or exchanged in the grid, and more on the solution of extreme cases, such as a partial failure of the grid or essential nodes [AG12]. They demonstrate their approach using a low-voltage grid in which a central node fails, and the system itself attempts to rectify this failure by setting switches in a self-organised manner. Agents, therefore, do not trade with each other but switch powerlines on or off to stabilise the grid in case of a fault.

Approaches such as the microgrid, as described here [LP04], are similar to the cellular approach in that they also divide the electricity grid into smaller subparts.

However, the aim behind this is different, as microgrids are designed to disconnect in the event of voltage deviations or other faults in the higher-level power grid and switch to island operation to continue operating the loads within the microgrid autonomously through local generation. Local generation can only be emergency generation, such as gas generators that switch on quickly, but also standard feed-in generators, such as the DRES. Internally, microgrids also implement a decentralised organisation through a P2P network to prevent the failure of central control units within the grid and assure autarky in every situation. Their chosen approach promotes the basic concept of consuming generated energy as regionally as possible. However, unlike the approach chosen in this thesis, they do not make any statements about the communication between the microgrids. They are, therefore, not a general approach to transforming the power grid into a cellular system, as they focus too much on the individual cell and ignore balancing the whole grid.

According to the review of related work, it can be shown that the current research work on decentralising and adapting the energy grid mainly focuses only ever on one of the two approaches: self-organisation or cellularity. In this thesis, however, we want to extend and modify the VDE approach of a cellular energy system with self-organisation to develop concrete implementation for decentralised and regional energy distribution in the power grid.

3. Methodology

Now that the necessary fundamental concepts have been introduced in section 2, an approach to the concrete implementation of a decentralised self-organised cellular energy system is outlined. This approach considers the quarter-hourly energy sharing with consideration of grid losses as a core functionality, other system services as reactive power and voltage control, need to be considered separately in future approaches. In addition, it is shown that the approach developed adheres to the requirements and characteristics of a self-organised system and considers and implements the basic concepts of the VDE cell. Firstly, several assumptions are made under which the concept was developed so that it can concentrate fully on the main focus of the research topic, namely the most regional energy trading possible in regular grid operation. This follows a rough introduction to the concept before going into more detail on the two main components: the cell as an agent and the interaction between them.

3.1. Assumptions

Firstly, in contrast to the VDE cell, our focus is solely on the electricity grid as an energy distribution grid. Therefore, the principle of sector coupling only plays a role in providing flexibility within a cell and not as an option for the alternative exchange of energy between cells. This focus is because electricity grids offer the least transport flexibility of all grids and, therefore, play the most critical role in grid balancing. A deviation from the control voltage in the electricity grid can lead to its immediate collapse, which is not the case in gas grids, for example. The VDE paper also sees the electricity grid as the core grid, supplemented by alternative energy sectors, like gas and heat [JB19].

Another assumption is that the network is in a "normal" state, i.e. the grid structure does not change during the trading process. All components, but above all the cells, behave consistently and with integrity and do not fail during trading.

Negative prices in the electricity grid, as they occur on the traditional electricity market, are also ignored. The assumption of exclusively positive prices (including free production) is valid, as negative prices on the current electricity market are mainly due to the inertia of fossil-fuelled power plants. Unlike DRES, many large power plants cannot simply be switched off in the event of overproduction in the grid,

3. Methodology

as restarting them would incur higher costs than selling the energy under penalty charges. They may also not be switched off for conceptual safety reasons, as is the case with nuclear power plants, for example. However, as our grid builds up mainly using DRES, this assumption is valid for us.

Regarding flexibility, we limit ourselves to the amount of energy that gets traded, as only the trade planning is considered. This implies that the ramp rate of flexible loads such as Power-2-X technologies or other loads is negligible. The ramp rates for energy generation can also be neglected, as DRES require a very short start-up time compared to the quarter-hourly trading slots.

In principle, the approach assumes that the load can only be covered in full or not at all. This assumption is based on the fact that some energy consumers can only be throttled with difficulty or not at all. Large industrial plants, such as steel smelters, require a constant level of energy and are long and expensive to start up. Partially covering the load would have no advantages for them, as the plant can only function with its entire load covered. In order to create a small common denominator for all consumers, we assume non-division load bids.

In contrast, we regard generation as division bids, as DRES can flexibly provide the required generation. Therefore, it is always worthwhile for them to provide the energy regardless of its amount.

Finally, we assume that there are only flexible loads in the electricity grid. Each load is assigned a maximum price, i.e., the maximum price that will be paid to cover the load. This price also assigns importance to each load, as higher prices mean that the load must be filled more urgently than with lower prices. An energy storage system that needs to be filled sporadically should only be topped up with cheap PV energy, while the industrial plant must always be running and is, therefore, given a higher price and greater importance. Even inflexible loads have a finite maximum price, above which it is more economical to disconnect the load from the grid than to pay to cover it. The maximum price results from a combination of mapping the importance of a load, the economic yield it can generate, and the average generation price in the neighbourhood. Let us, therefore, assume that a load has an almost infinite maximum price. In this case, this indicates a lack of generation in the neighbourhood and should be an incentive for new generation sites in the neighbourhood. Therefore, the maximum price for the load is naturally capped by the market economy, making this assumption valid.

3.2. Self-Organized energy-sharing among energy cells

The basic concept for self-organised regional energy sharing is identical to the VDE concept. This approach also adopts a bottom-up approach in which the grid is

3.2. Self-Organized energy-sharing among energy cells

divided into self-similar cells at different cell levels. At the lowest level, a cell aggregates producers and consumers. The higher the cell level, cells remain self-similar but can now include other cells besides consumers and producers, as seen in Figure 3.1. For example, the cell marked in grey on View 1 encloses four other cells from the cell level below. Therefore, the entire cellular energy system is made up of nested cells. In our approach, agents represent cells. This also implicitly guarantees the self-similarity of the cells, as the required autonomy for agents means that there can be no administratively higher-order cells across all cell levels.

The principle of subsidiarity requires that energy balancing or trading occur first at the respective cell level. In Figure 3.1, this would mean, for example, that the four cells on View 2 first balance their load among themselves as far as possible before contacting the higher cell level, similar to View 3.

In order to emphasise the principle of subsidiarity, our approach narrows its focus to a single cell level from this point onwards, as indicated in colour in Figure 3.1. This implies that the subsequent energy trading concepts solely pertain to the interaction between agents at one cell level.

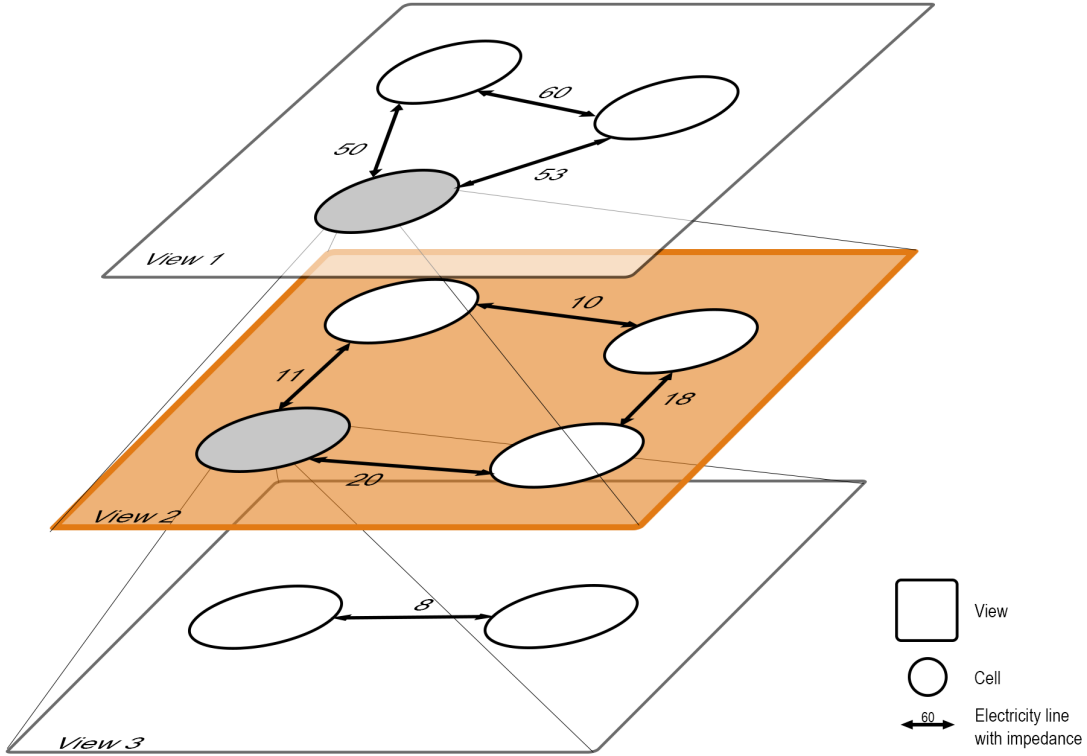


Figure 3.1.: High-level System architecture of the developed cellular energy grid

Power lines connect cells. These lines have a power loss or line resistance, represented by the impedance, see Figure 3.1. In energy trading, the model then adds this impedance to the producer price in the form of transport costs to consider grid

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losses. This forces the power generated to be consumed as regionally as possible. It should be noted that the transport costs depend proportionally on the line resistance, i.e. thick high-voltage lines have lower transport costs than lines with a higher resistance, as found in low-voltage grids. Considering that the length of the power line also influences the impedance (see subsection 2.1.3), the chosen approach also reduces the probability that cheap energy is transported over long distances to its destination, regardless of the local generation rate, as is currently the case in the electricity market using north-south routes. This and the fact that agents always aim to cover their current loads as cheaply as possible fulfils the regionality of electricity consumption, one of the main objectives of the research question.

Now that the energy system and its basic concept have been roughly introduced, we will focus more closely on the agents, their requirements, and, subsequently, their interactions.

3.2.1. Agent: The Cell

As previously stated, each cell in our self-organised system functions as an agent and vice versa. The approach implements a weak self-organised system, as the agents each have an internal central control unit, also known as an ECM. The internal central control facilitates local data aggregation, like demand and supply within the cell, for predictions and provides a central interface to all neighbours through the ECM, which is required in the VDE approach [JB19].

Moreover, each cell, and consequently the ECM, is designed to only be aware of its direct neighbours, including their line resistances. This design choice ensures that the ECM can only interact locally, a key characteristic of a self-organising system. For instance, if a cell needs to communicate with neighbours of neighbours to receive a supply offer, the interaction must occur via hops, i.e. a direct neighbour, which in turn is aware of the line resistances to its neighbours.

In addition to the exclusively local interactions, cells have exclusively local goals. One of these is that agents want to cover their loads as cheaply as possible. Another one is that agents, regardless of their state (balanced, undersupplied or oversupplied), always aim to cover their loads.

Through local interaction and local goals, agents pursue global emergent behaviour, which occurs in self-organised systems. The emergence reflects a global energy transfer that ends in equilibrium, i.e., in a broadly balanced grid.

One of an agent's main tasks is trading energy with others. Thereby, the principle of solidarity must always be observed. This means that cells are obliged to help each other with energy equalisation and prohibits any speculative, market-based decision in favour of one's own cell.

Before the actual energy trading can start, the agent enters the pre-trading phase. In this phase, the ECM forecasts the expected load for the future trading slot in conjunction with a maximum price. This is the last price the agent would pay before

switching off the load would be more economical. The same applies to the generation forecast, although this is not linked to a maximum price but to the generation price. Different prices for loads and generation express their different nature or importance, e.g. electricity generation from PV is cheaper than from biogas, and an industrial load may be more important than charging a storage facility. At the time of trading, each agent has, therefore, a set of load-price pairs and a set of generation-price pairs. Each agent must cover its loads in order of importance, i.e. loads with a higher maximum price must be covered first. If the ECM's prediction does not materialise, the cell is forced to switch off its load and pay a penalty.

3.2.2. Interaction: The Trade

After each agent has made the necessary preparations, energy trading starts for each agent at a globally defined time. This consists of two states, as seen from the state machine in Figure 3.2. In the initial state, also known as the explore state, each agent searches for potential trading partners, considering grid losses. In the subsequent accept state, each agent then tries to cover as much of its load as possible with the help of the trading partners found.

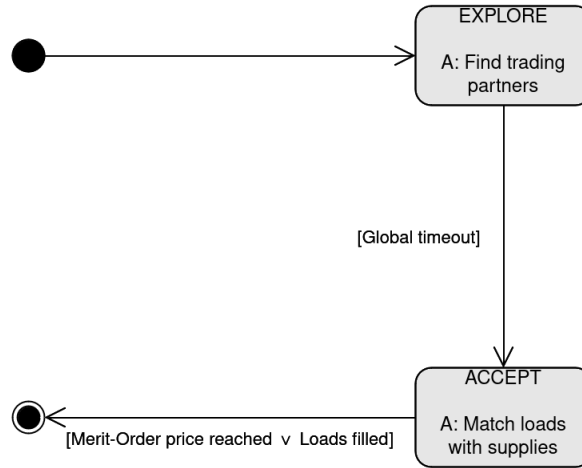


Figure 3.2.: State diagram of the energy trade

The state machine in Figure 3.2 shows both states and their corresponding described actions. It also shows the conditions for the transition to a new state or the end of energy trading, which are shown in square brackets.

Explore - State

As briefly mentioned above, the agent determines its visibility in the network in the explore phase, i.e., how far it can interact with neighbours of neighbours to

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ask for supply offers for its loads. The exploration is similar to the principle of Dijkstra's algorithm from graph theory. Agents or cells represent the nodes, while edges indicate whether a power line exists between the cells and what impedance it carries (see Figure 3.3). As a result, the exploration, at the end of the explore state, spans a tree that encompasses all the shortest paths between the agent and its potential trading partners. Marked by orange edges in Figure 3.3.

To prevent the exploration from traversing the entire grid globally, the upward visibility is limited by the following inequality:

$$cost_{transport}(shortest\ path) \leq max(load\ prices)$$

This means that as soon as the transport costs of the shortest path exceed the highest price that the agent is willing to pay for its loads (see section 3.2.1, last paragraph), the visibility limit in this direction is reached. No further trading partners are then sought in the same direction, i.e. via this path. We call this reaching the trading cap for this direction. Reaching the trading cap for all directions means the exploration found all possible trading partners for the agent. The agent has reached its local trading cap.

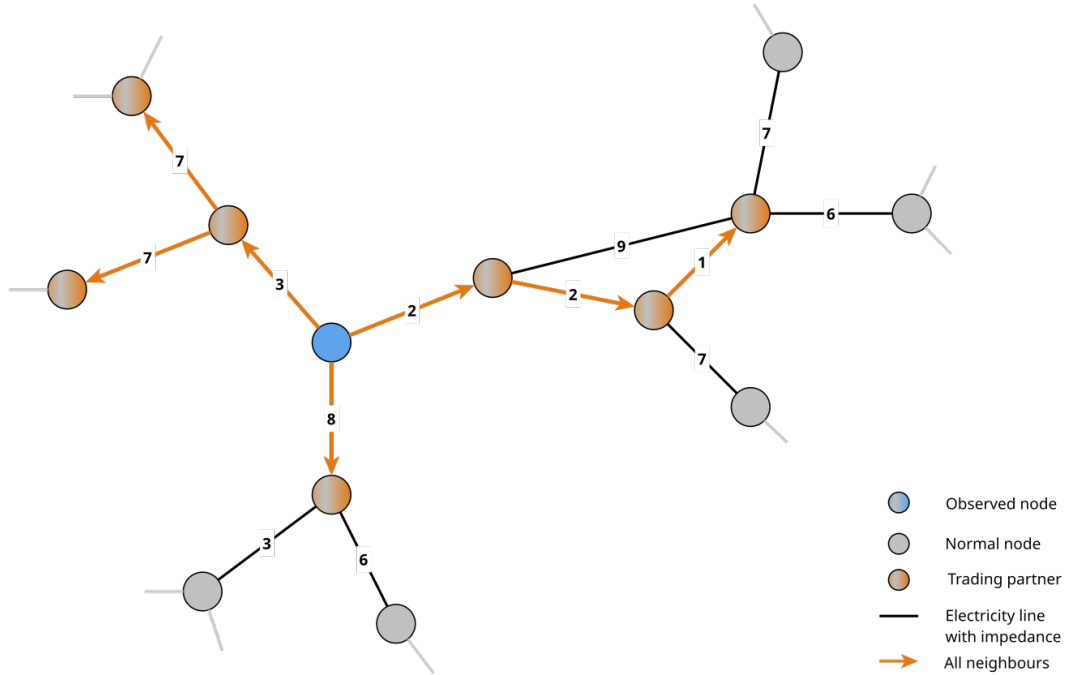


Figure 3.3.: Stretched subtree at the end of the explore state with trading cap at 10

In Figure 3.3, we make the simplified assumption that the maximum price for the loads of the agent marked in blue is ten units. This results in the orange-coloured

tree with path weights corresponding to a maximum of ten units. The grey-coloured nodes correspond to the agents outside the visibility range of the blue agent. Except for itself, the blue agent can, therefore, call on seven trading partners in order to cover its loads. While searching for trading partners, the agent simultaneously calls up all supply offers from the agents found and saves them for the next state. Therefore, each agent stores a set of tuples (*energy amount, price, transport costs*) from all possible supply offers of the trading partners. At the end of the Explore state, the blue agent in Figure 3.3, therefore, knows all the trading partners and all their supply offers.

Two essential aspects characterise the chosen approach to exploration and, thus, the entire energy trade:

Heterarchy becomes hierarchy: Each agent builds a tree based on a part of the heterarchical network, i.e., it builds a partial hierarchy containing its visible trading partners (see Figure 3.3). This overlay network, emphasised by directed edges in Figure 3.3, describes the communication paths between the agents. However, it also assigns minimum transport costs to energy trading between the agents, which must be taken into account when trading.

Minimale Transportkosten: In contrast to routing on the Internet, electricity does not just flow from the source to its destination via a single path. It always chooses the path of least current resistance. This implies that a large part of the current flow will choose lines with low impedance, but a non-negligible part will also flow via lines with higher impedance. Similar to the flow behaviour of water in a branched channel system. In order to find an exact value for the impedance and, thus, the transport costs for trade between agents, a power flow analysis would be required. This is explanatorily explained here [CCH⁺91]. Since our approach focuses on planning regional energy distribution and not the exact calculation of line loads, we are satisfied with the lower bound of the possible path impedances, meaning the minimum transport costs. They are sufficient to quantitatively take grid loss into account in our model, as they order trading partner ratio scaled according to the transport costs.

In the Accept state, energy balancing occurs based on this simplification, considering the minimum transport costs as grid losses. Whenever transport costs are mentioned in the following, this automatically refers to the minimum transport costs.

State transition

After the end of the exploration phase, the state changes to the "Accept" state. This transition can take place under several conditions. The three most prominent ones are described below, along with their possible problems.

- **Based on the local trading caps:**

Each agent changes state independently after reaching its local trading cap. The state change, therefore, takes place asynchronously. This means that different agents maintain different states throughout the energy trade. For example, an agent with few trading partners falls into the "Accept" state very quickly because of high outgoing impedances. In contrast, agents with many trading partners take much longer to change their state. As a result, the approach becomes unfair, as the latter do not have an equal chance of receiving cheap supply offers. Rejecting this approach is, therefore, the only viable option.

- **Based on a global trading cap:**

One solution to the above problem would be introducing a global trading cap. It is reached when all agents arrive at their local trading cap. Reaching the global trading cap triggers a state change automatically, and all agents start negotiating synchronously. Two significant problems make this approach impractical. Firstly, a global trading cap requires global control or communication between the agents, which self-organisation excludes explicitly. Second, a single agent globally traversing the network would block the entire system for a long time. Both points prevent the realisation.

- **Based on a synchronous timeout:**

A better, ultimately implemented approach is to use a global synchronous timeout. As soon as the timeout is reached, all agents must perform a synchronised state change regardless of whether their local trading cap is reached. Thus, similar to the above approach, the agents start negotiating synchronously. This approach also solves both problems of the global trading cap. On the one hand, global interaction or control is no longer necessary as the timeout is static information. Secondly, a globally traversing agent can no longer block the entire process. The determination of a good timeout is crucial and is best achieved through real-time simulations of the grid.

In this approach, the global timeout is also the preferred method. This is due to its practical implementation and compatibility with the principles of self-organisation. It also ensures fair behaviour between the agents, as shown in Figure 3.2 in the transition between the two states (square brackets).

Accept - State

Now that each agent has all possible supply offers from their visible trading partners, it is up to them to cover their load with the help of their supplies and the supply offers they have collected. In principle, there is no difference between their own supply and the supply of their trading partners. This means that their supply is not favoured or used first. Only the associated transport costs differ. While their own supply has none, the supply offers of their trading partners are linked to the

transport costs described above. During the accept phase, the algorithm undertakes the task of matching loads (including their associated maximum prices) with the available supply offers. This matching process has two key requirements. Firstly, the total amount of the loads must be covered by one or more supply offers. Secondly, the sum of the supply offer costs, including their transport costs, multiplied by their respective energy amount must not exceed the product of the maximum price (set for the respective loads) and its according load amount. This describes the the following inequality which must always be met during the matching process:

$$amount_{load} * maxprice_{load} \geq \sum_{i=1}^n amount_i * (cost_{supply,i} + cost_{transport,i})$$

Where n is the number of supplies required to cover the respective load.

In line with the assumptions from section 3.1, the energy trading approach guarantees that loads are always covered in full or not at all. This means covering only half or part of a load with supply offers is impossible.

In the following, when using the term supply costs, their associated transport costs are also implicitly included for better understanding. The following assignment therefore applies: $supply\ costs := cost_{supply} + cost_{transport}$

There are two general approaches for matching loads and supplies:

- **Matching using the knapsack problem:**

In line with the Knapsack problem [KPP04], the aim is to cover as much load as possible with the lowest possible supply costs. This means that cheap supply offers or combinations of them are only used for loads with a low maximum price. Meanwhile, loads with a high maximum price are only covered by more expensive supply costs. The aim is to cover an optimum load with combinations of the available supply offers. The challenge here is that loads cannot only be partially covered. This optimisation problem precisely describes the NP-hard knapsack problem in its general form. This matching approach can be chosen if the focus of energy trading is on optimised load coverage, regardless of the significance of the maximum price of a load. However, since this price also implicitly encodes the importance of a load and every agent must cover its loads according to this scale of importance, a different approach must be chosen since the knapsack problem cannot guarantee this ranking.

- **Matching using a modified Merit-Order principle:**

The cheapest supply costs always cover the most expensive load. This results in two intersecting curves, as the supply costs generally increase while the maximum load prices decrease. The increasing supply costs result from rising transport costs as soon as regional supply offers are used up. However, the maximum prices for the loads will decrease due to important loads already being covered, and there are now less important, i.e. less expensive, loads to fill. Each agent, therefore, calculates its local merit order price, which is the intersection of the two curves. As soon as this point has been reached or all

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loads have already been covered, the algorithm stops, as the above inequality can no longer be fulfilled. This marks the end of load balancing and, at the same time, the end of energy trading for this trading slot (see Figure 3.2). The calculated local clearing price forms the basis for the actual exchange price. This means that each agent pays exactly this price at the end for all loads included in the merit order process. Similar to the merit-order principle in the centralised electricity market.

The latter approach, which we have implemented as our decision criterion, offers significant benefits in the context of energy trading. It not only ensures that the importance of the loads is duly considered but also guarantees that load balancing is always as regional as possible. This is particularly advantageous in situations where rising transport costs could increase the total costs due to a lack of local offers. The merit order approach, however, strives to keep these costs as low as possible by always accepting the cheapest offer and, therefore, balancing as regional as possible. Suppose transport costs permit and energy from more distant trading partners is very cheap. In that case, the approach also promotes using the cheapest energy first, i.e. potentially the more sustainable energy (see section 2.1.4).

Offers are always inclusive (solidarity principle), which is why there may be multiple requests for a single supply offer. The requested agent must resolve this conflict. It is also possible that trading partners only want to call up part of the offered supply. The supply offer may have to be split up. In principle, it is possible that parts of the supply offer flow to different trading partners or that a residual amount is not taken. Offer splitting on the producer side is explicitly excluded. This means that an agent does not split its supply offer among the trading partners who request it. This is intended to avoid excessive fragmentation of supply offers.

There are two approaches to resolving multiple requests:

- **Asynchronously:**

On the one hand, an asynchronous approach based on the first come, first serve principle can be used. Accordingly, the agent who first makes a binding request for the load receives the offer. If the agent only requests part of the load, the remaining amount is available for the other agents. The advantage of this approach is that there is no need for synchronisation between the agents. If a requesting agent does not receive its offer because it no longer exists, it must try to cover its load with other offers.

- **Synchronously:**

On the other hand, a synchronised approach can be chosen. For example, the agent with the lowest transport costs always gets the contract. Alternatively, there could be an auction among the requesters for the respective supply offer. Both approaches mean that the agents have to synchronise with each other, which can lead to a possible deadlock. This risk must be analysed in more detail in further work.

The trading algorithm implements the asynchronous approach, which ensures simple communication between the agents and avoids a synchronisation overhead. Figure 3.4 shows the chosen approach based on the three response options: Offer is still available, Offer is only partially available, and Offer no longer exists. In the first two cases, an acknowledgement is sent, whereas in the second case, the remaining amount must also be communicated. In the latter case, a negative acknowledgement is sent. This communication does not require synchronisation between the agents.

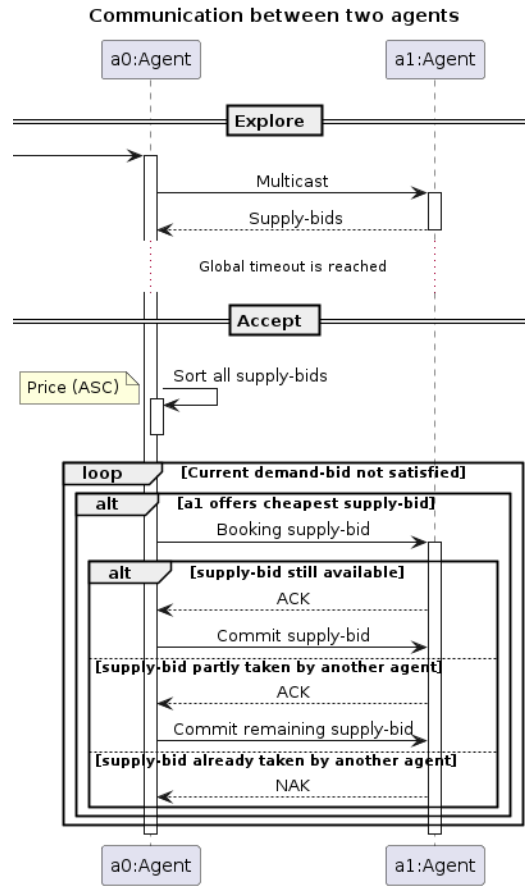


Figure 3.4.: Information exchange between two direct neighbours during the whole energy trade

To summarise, the sequence diagram in Figure 3.4 shows all the steps described in their order. Energy trading starts with each agent sending a multicast to its neighbours to determine its range of vision in the network. At the same time, it also receives the supply offers from each trading partner it finds through individual supply bids. When the state change occurs, the agent sorts the received supply bids according to their costs. These are the sum of the specified production costs and the transport costs incurred. In the Accept state, the agent then covers its loads with the cheapest supply bids, according to the modified merit order principle. Figure

3. Methodology

3.4 assumes that the direct neighbour agent a_1 has submitted the cheapest supply offer, so agent a_0 tries to book the bid with him. In response, he received the answer options that were already explained above. If a_0 cannot book the offer because it is no longer available, he must choose another one. As soon as the entire load has been covered or the maximum price for the loads is exceeded, energy trading ends, and the agent prepares for a new trading slot. If loads remain open after energy trading, they must stay switched off.

Now that the functioning of energy trading has been explained, the most essential feature of the approach, regionality, will be discussed. Energy should first be consumed locally or regionally before offers with higher grid losses are accepted. This requirement is implemented through both phases of energy trading. Firstly, the local trading cap limits the number of available agents locally. Subsequently, the modified merit order principle ensures the most regional selection possible from the already limited number of agents available for trading. This promotes regionality, helps make the approach scalable, and adheres to the principles of self-organisation. However, a natural worst-case scenario exists for the energy trading system previously described. This occurs when the maximum price of a load exceeds the transport costs of the grid diameter. In such instances, the local trading cap can't prevent global traversal of the grid. Here, the modified merit order price steps in to maintain regionality. If it also fails to limit the trading behaviour due to the local generation being too expensive or unavailable, the high transport costs automatically drive up the merit order price. This leads to an economic incentive to locate flexibility or generation plants in the neighbourhood in order to take advantage of the high merit order price of the cell. A global traversal of the grid should therefore only be a temporary event.

4. Evaluation and Discussion

Now that the concept of self-organised energy trading, considering grid losses, has been presented in the cellular energy system, it is time to simulate this approach and evaluate it using a realistic example. Additionally, questions regarding the algorithm’s convergence and regionality are discussed.

For this purpose, uniformly distributed energy data will be tested on real power grid topologies. Firstly, the approach is simulated on a meshed extra-high voltage grid (EHV). Subsequently, topology differences are analysed between the discussed EHV grid and a semi-open radial network, a medium-voltage grid (MV).

4.1. Experimental Setup

Initially, the basic situation and the input parameters under which the simulations took place are described. All simulations were always carried out on the extra-high voltage grid (EHV) or the medium voltage grid (MV). Both originate from the comprehensive SimBench testbench collection [MSD⁺20], which provides realistic scenarios for different grid types. Cells took on the role of buses, and lines were adopted with minor adjustments. The test grids used can also be seen in Figure 4.1.

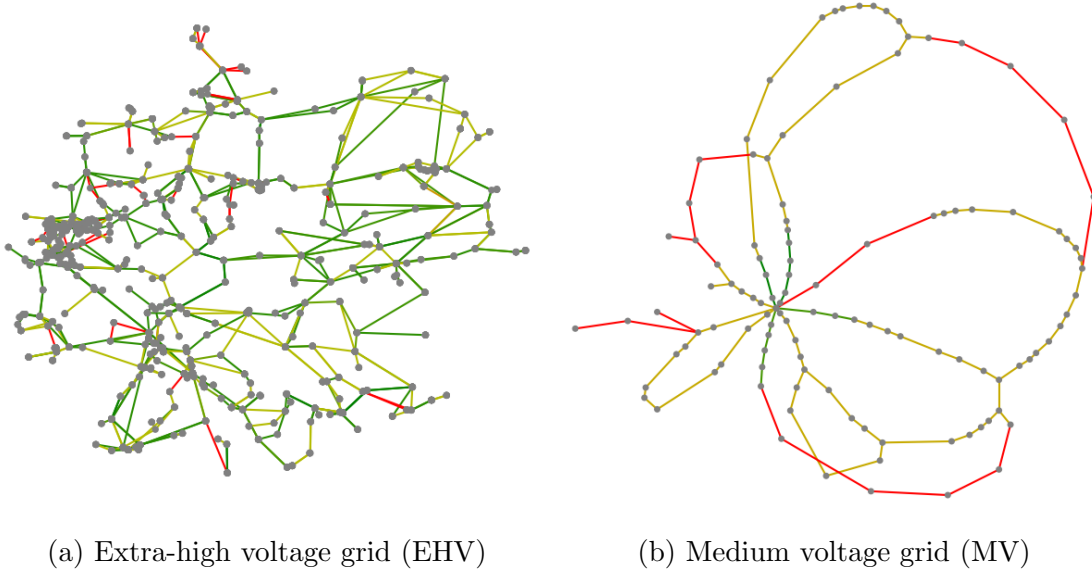


Figure 4.1.: Example power grids with line impedance ratio

4. Evaluation and Discussion

The EHV network (Figure 4.1a) is characterised by its meshing and, above all, by its predominantly low line losses. The colour scale from green to red describes the impedance level on the respective line, with low impedance marked in green.

On the other hand, the MV grid (Figure 4.1b) presents a different picture. It is more radially connected, and its line losses tend to be concentrated in the upper spectrum. These differences in the topology can also be analysed using the clustering coefficient from Table 4.1. It describes the extent to which the network nodes are connected to form a clique. Although it is also low in the EHV grid due to its structure, it is still significantly higher than in the MV network, in which there is no clique formation and, therefore, no meshing. The ratio of lines to cells also indicates the different topology. For example, the EHV grid has an average of 1.3 cells per line, while in the MV grid, there is almost a one-to-one mapping (see Table 4.1).

Attributes	EHV	MV
Agents (Cells)	571	115
Lines	752	122
Diameter (Transport costs)	353	890
Clustering Coefficient	0.010085	0.000000

Table 4.1.: Attributes of the grids

The low impedances and correspondingly low transport costs in the EHV grid can also be seen in Figure 4.2. The boxplot shows that most transport costs are between two and 15 units. The average degree of a cell (see Figure 4.3) is also low. This means no high transport costs must be overcome to trade with direct neighbours. Even more distant trade is not overly restricted by high transport costs, as the diameter (see Table 4.1) of the EHV network is comparatively tiny at 353 units, considering the grid's size.

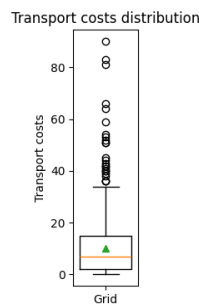


Figure 4.2.: Transport cost distribution of the EHV grid

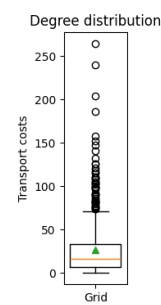


Figure 4.3.: Degree distribution of the EHV grid

The transport costs in the MV network are significantly higher, which can be com-

pared using Figure 4.4. The boxplot shows that most costs are between 18 and 55 units. This also influences the degree of a cell in the medium-voltage grid, as seen in Figure 4.5. It is more than twice as high as in the EHV grid. At 890 units, the diameter is also approx. 2.5 times higher than in the EHV grid. Overall, energy trading in the MV network will result in significantly higher transport costs.

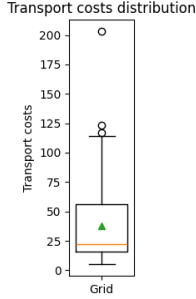


Figure 4.4.: Transport cost distribution of the MV grid

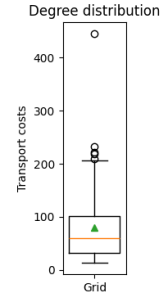


Figure 4.5.: Degree distribution of the MV grid

In order to keep the verification of the developed approach as trivial as possible and to analyse possible effects of the structure of the grids on the developed energy trade, an even distribution of generation and consumption, as well as their prices, was assumed, regardless of the grid type. Table 4.2 shows which parameters were used to generate the distributions. After initialisation, each agent has a random number of loads (load-bids) and supplies (supply-bids). The possible minimum generation or load was set at ten MW to prevent trading in unrealistic minimum quantities. The distributions imply that a balanced grid was considered in the simulation. This is because permanently unbalanced grids should generally be balanced by higher-level grids i.e. cells, or there is an economic incentive to balance the grid through installing new suppliers.

Agent parameter	Result space
Load per bid	indep. unif., $\Omega = [10, 100]$
Price per load	indep. unif., $\Omega = [1, 100]$
Supply per bid	indep. unif., $\Omega = [10, 100]$
Price per supply	indep. unif., $\Omega = [1, 100]$
Number of load bids	indep. unif., $\Omega = [1, 10]$
Number of supply bids	indep. unif., $\Omega = [1, 10]$

Table 4.2.: Attributes of the agents

Agent-based modelling and simulation (ABMS) was used to simulate the developed approach. Accordingly to the use case, the network topology of ABMS was chosen to implement the neighbourhood identically to the grids. Using ABMS, the model

4. Evaluation and Discussion

was simulated under nine different combinations of random variables for generation and consumption and then totalled and averaged across all results. This was necessary to obtain reliable and stable results and include a possible rebound of the flexible loads.

Furthermore, the asynchronous nature of trading (see Section 3.2.2) was simulated using a random variable. This variable affects the order in which the model executes the agents. The results were totalled and averaged across all grid agents so that reliable results can be assumed.

Identical to the assumptions in Section 3.1, the model also simulates non-division bids on the consumer side and division bids on the producer side. The ABMS only runs a single trading step, i.e. 15 minutes of energy trading, to focus on and verify the functioning of the actual energy trading. The Mesa Framework [KMC20] was used and extended by a network topology to implement the simulation. All evaluation results and the developed simulation model can be viewed on the GitLab instance of the Chair of Computer Networks and Communication at the University of Passau [See24].

As already stated, the approach’s primary verification is only carried out using the EHV grid to avoid an unnecessary, redundant listing of identical observations from the MV network. Therefore, the results and figures in the following two sections are only taken from the simulation of the meshed grid.

To provide a better overview, the caption of each figure nevertheless states which grid type it belongs to. EHV therefore stands for the simulation of the extra-high-voltage grid and MV for the simulation of the medium-voltage grid.

4.2. Basic energy trading procedure

The entire energy trade takes 31 trading steps, which can be seen in figure 4.6. Of these, 25 steps relate to the Explore phase and six to the Accept phase of the two-phase energy trading process. Actual energy exchange only occurs in the latter phase, as the potential trading partners must first be determined in the Explore phase.

Figure 4.6 shows the remaining supply across the entire energy trade. It can be seen that approximately half of the supply gets traded at the end of energy trading. The consumed generation is equivalent to the covered load as the load and generation are equally distributed in the simulation. This means that half of all loads incurred are covered at the end of the energy trading period.

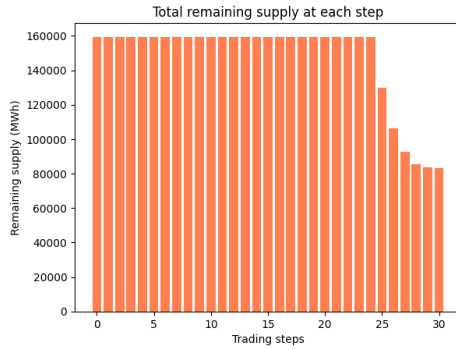


Figure 4.6.: Total remaining supply (EHV)

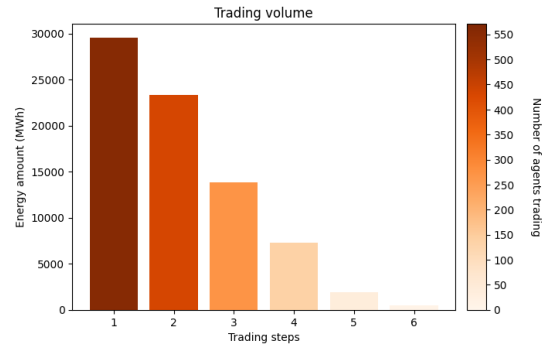


Figure 4.7.: Decreasing trading volumen (EHV)

The largest trading volume and, therefore, the largest load coverage occurs in the first trading step of the Accept phase at 29573 MW. This coverage decreases exponentially over the trading steps until the model converges at 83011 MW of remaining load. This can also be seen in Figure 4.7, which shows the decrease in trading volume and trading agents.

The decrease with subsequent convergence could be due to the decreasing maximum price for the load. Agents can no longer cover their loads with the available supplies as this is impossible with the given price. Figure 4.8 shows this decrease in the maximum price of the load and the associated decrease in the importance of covering this load. On average, loads with a maximum price of 58 units are still covered at the end of energy trading in step six. It also shows the lower limit of the maximum price for a load that can still be filled. There are still cheap offers in the first trading steps, so the limit is initially very low. As the trading steps increase and these offers disappear, the boundary rises to 43 units. This means that loads with a lower maximum price are no longer covered.

The Accept phase also stops after six individual trading steps, as the load in the seventh trading step can no longer be filled under the low maximum costs. This means that only a maximum of six load bids, i.e. six independent loads per agent,

4. Evaluation and Discussion

are covered. The maximum prices would have to be set higher to cover more load in total.

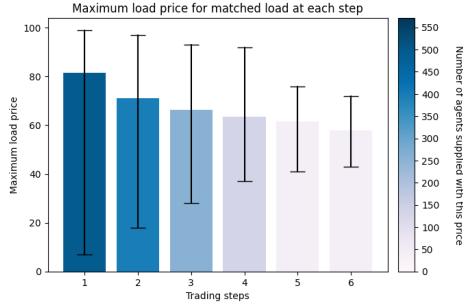


Figure 4.8.: Maximum price for matched loads (EHV)

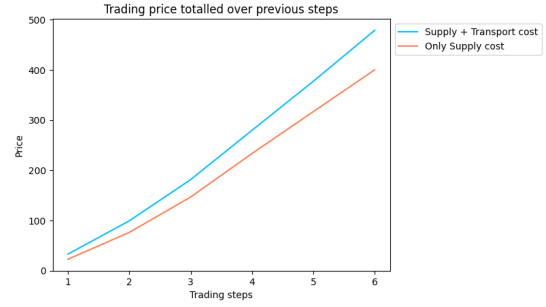


Figure 4.9.: Trading cost totalled over the previous steps (EHV)

However, the decrease in trading volume is also due to the local sell-off of cheap supplies. Important loads must be covered by distant cheaper or more expensive local production. This automatically drives up the total costs (supply costs + transport costs) for trading and leads to the maximum price possibly being exceeded. Figure 4.9 clearly shows the increase in transport costs. It shows the supply costs with and without transport costs totalled over the previous trading steps. As the number of trading steps increases, both curves diverge and show that transport costs are playing an increasingly important role. At the same time, the slow divergence also shows that only necessary transport costs are accepted and that cheap offers with high transport costs are not sought globally.

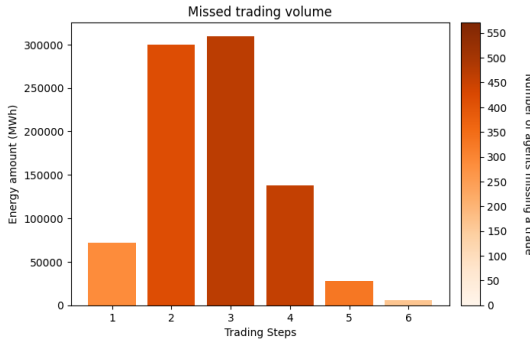


Figure 4.10.: Missed trading volume (EHV)



Figure 4.11.: Number of agents missing a trade (EHV)

Figure 4.10 and Figure 4.11 further underline that agents must enquire about more distant offers with increasing trading steps. While minimal supply volume has already been used up in the first step of the Accept phase, the bar shoots up significantly in steps two and three. Agents have outdated local offers in their list of

available supplies that they want to use to cover their current load.

As the offers have already been consumed, they must find new supplies. This missed supply volume decreases again over the following steps (four to six) as the number of agents still trading at all decreases (orange curve in Figure 4.12).

Figure 4.11 also shows an unwanted communication overhead, as agents communicate with each other over outdated offers, causing unnecessary interactions. However, this is difficult to avoid due to the enforced abstinence of global knowledge.



Figure 4.12.: Decrease of the number of agents and trades over all trading steps (EHV)

The decrease in trades and trading agents in relation to each other can be seen in figure 4.12. While many agents initially trade a lot, both the number of trades and the number of trading agents decrease almost simultaneously.

Initially, 556 agents were responsible for 1034 trades. After trading step four, there are 138 agents for 271 trades. This parallel convergence of both curves shows that although the number of agents is decreasing, the remaining ones, which still trade, continue to cover their loads with an approximately equal number of trades.

4.3. Visibility range and matching process

Now that basic observations on energy trading and its effects have been made, the two central aspects of the novel approach will be discussed in more detail. These are the local trading cap as a visibility criterion and the modified merit order price as a decentralised matching procedure. Both are also only discussed on the results of the EHV grid.

Before the actual energy exchange takes place, the local trading cap must first define the visibility range of each agent in the Explore phase. Figure 4.13 shows that the average local trading cap is reached after transport costs of 78 units. This means that, on average, agents can traverse a maximum of 22% of the graph diameter to

4. Evaluation and Discussion

find their trading partner and trade with them in the next state. In the worst case, a maximum of 28% of the graph diameter is possible.

The directional trading cap is significantly lower, with an average maximum transport cost of 62 units (see Figure 4.14). This difference is because the local trading cap is an agent's maximum directional trading cap. In other words, the upper bound of all agent's direction trading caps. This shows that, on average, agents could only traverse a maximum of 18% of the graph diameter in one direction. The local trading cap, therefore, only represents the maximum worst-case traversal.

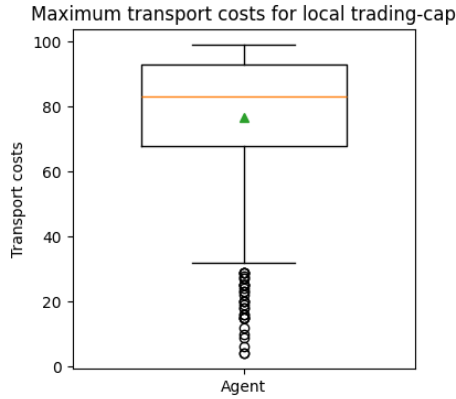


Figure 4.13.: Local trading cap (EHV)

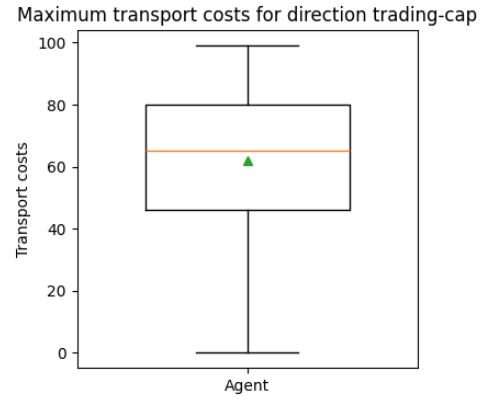


Figure 4.14.: Direction trading cap (EHV)

However, the actual trading distance is primarily determined by the modified merit order principle in the Accept phase and is only limited upwards by the local trading cap.

On average, according to Figure 4.15, agents incur transport costs of approximately six units across the first four trading steps. This means that they trade with their direct neighbours. With increasing trading steps, the agents have to look for more distant offers, as mentioned above, which is why the transport costs rise to ten units. However, this average upper limit is only reached by few trades (see colour bar). Figure 4.12 shows that a small amount of trades also mean few agents still trading. This provides the insight that only a minor amount of agents reach this upper limit. The rest exited energy trading earlier due to reaching the merit order price. Figure 4.16 underpins this observation. It shows the progression of when agents reach their local clearing price and thus cease energy trading. 90% of all agents reach it within the first four trading steps, most of them in step two.

On average, most agents trade with their immediate environment before they reach the local clearing price. A small proportion can delay reaching the local clearing price by trading further away. These agents define, for example, low transport costs in the neighbourhood (cells from the lower quantile in Figure 4.3) or that they lie within the confidence interval of the maximum prices in steps five to six in Figure 4.8. Congruent with the above statement, the actual transport costs in figure 4.15

only start increasing for these remaining agents from trading step five onwards by an average of two units per additional trading step.

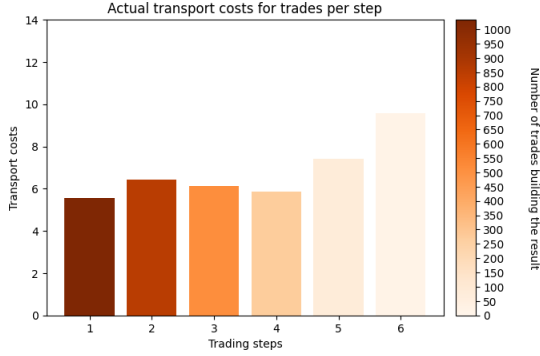


Figure 4.15.: Actual average transport costs (EHV)



Figure 4.16.: Agents reaching the local clearing price (EHV)

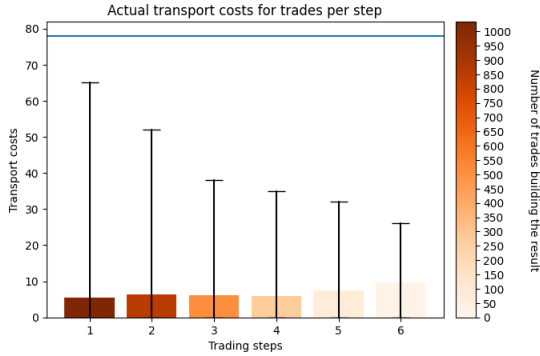


Figure 4.17.: Actual average transport costs with minimum and maximum values (EHV)

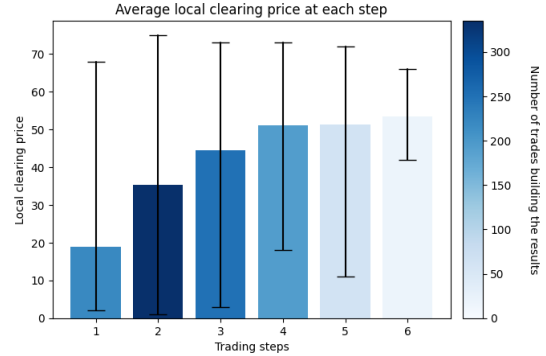


Figure 4.18.: Average local clearing price (EHV)

Figure 4.17 shows the maximum transport costs that can possibly occur. In some cases, these are around ten times higher than the average transport costs and only highlight one or few outliers. These absolute exceptions nevertheless define the maximum transport costs occurring in the model.

They reach their maximum at around 65 units and are well below the local trading cap. This means that although transport costs of 86 units would be possible in the worst case due to the local trading cap, the actual maximum trading range is below this and, therefore, remains even more regional.

Furthermore, the Figure shows the maximum transport cost incurred in the first trading step. This initially appears to contradict the above observation. However, these are only individual outliers. It means that although most agents in the first trading steps use their local supply or that of their direct neighbour, some agents

4. Evaluation and Discussion

accept a distant supply despite having their own available supply.

This happens if the distant supply is cheaper overall than their supply. With increasing trading steps, the possibility of very high outliers is pushed down by the falling maximum price for loads (see Figure 4.8). This is also why the maximum outlier lies in trading step one.

If we focus a little more on the agents' local clearing price, Figure 4.16 shows that around 20% of all agents reach their merit order price after covering the first load. This can be mapped to the fact that some agents assign a low maximum price to their loads. Initially, the load can still be filled during the first trading step. However, as the loads are sorted in descending order according to their maximum price, a further load with a lower maximum price can no longer be covered. To change this, the agent would have to price its loads differently.

Figure 4.18 also shows that the local clearing price rises with increasing trading steps. This increase is due to rising transport and production costs, as the cheap supply offers have already been consumed. Agents, therefore, have to accept more expensive trading costs to cover their loads or stop trading by reaching the local clearing price.

The maximum price of the loads causes the convergence of the local clearing price at approximately 55 units. This means that 55 units mark the maximum trading costs (supply + transport costs) for covering a load.

4.4. Impact of the grid structure

The final part of the evaluation presents differences between trading in the EHV grid and the MV grid. Both simulations were carried out with the same input parameters; only the grid type and the associated transport costs differed. This allows conclusions to be drawn about the influence of the grid structure on the trading approach.

The comparatively higher transport costs in the MV network (see Figure 4.4) mean that it is more expensive for agents to trade with neighbours. This is also shown in Figure 4.19. Although the average transport costs here are close to the EHV grid at a bit less than seven units, compared with the higher transport costs on a single line, it quickly becomes apparent that agents are more likely to be forced to trade locally with themselves.

Furthermore, unlike the EHV grid, transport costs do not increase towards the end of energy trading. With increasing trading steps, trading is discontinued due to excessive costs rather than extended to remote trading partners. The agents, therefore, automatically reach their local clearing price earlier. Compare the normalised curves from Figure 4.21.

Another key factor contributing to the more localised energy balancing is the chain-like structure of the MV grid. This structure leads to fast consumption of regional

cheap offers by direct neighbours. To access more such offers, agents would need to trade along the chain, which in turn results in steeply rising transport costs. Consequently, agents are compelled to cover their own load.

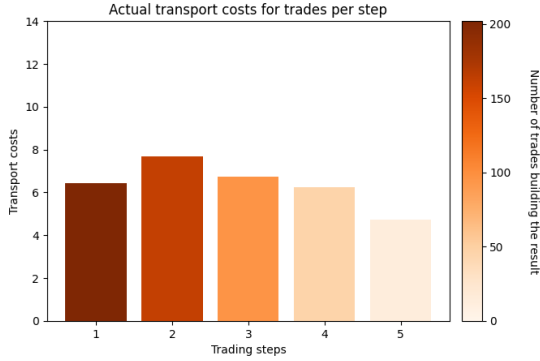


Figure 4.19.: Actual average transport costs (MV)

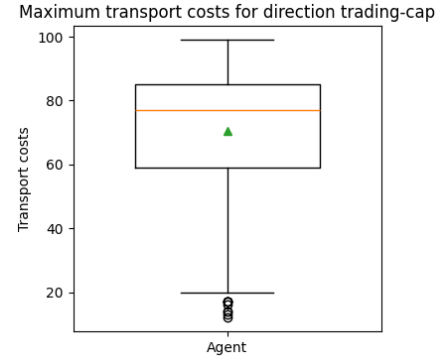


Figure 4.20.: Direction trading cap (MV)

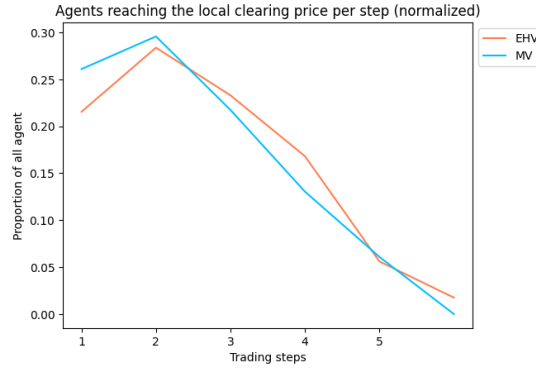


Figure 4.21.: Proportion of agents reaching the local clearing price

The structure of the two networks plays a crucial role in shaping the agents' visibility range as well. As the upper limit of all directional trading caps, the local trading cap remains consistent across the different network types. However, the directional trading cap is significantly influenced by the structure.

For instance, in the EHV network, half of all trading caps are confined between transport costs of 43 units and 80 units (see quantile interval in Figure 4.14). This range is notably narrower in the MV network, where half of the trading caps are limited between transport costs of 60 and 85 units (see quantile interval in Figure 4.20).

This can be attributed to the different meshing of the grids. Due to this, the EHV network has many possible directions and, therefore, many paths. Paths have different transport costs or lengths, resulting in a wide range of possible maximum costs for the directional trading cap. In the MV network, on the other hand, there

4. Evaluation and Discussion

are usually only individual long paths with high transport costs. This means that from the view of one agent, there are two possible directions. Therefore, the range of different sight distances in the respective directions is limited.

However, higher average transport costs for the direction trading cap and, thus, the maximum visibility in one direction do not mean that agents can find more trading partners in the MV grid. On the contrary, the diameter of the MV grid is much higher than in the EHV network. This means that agents, regarding the local trading cap, can only cross a maximum of nearly nine per cent of the grid diameter to find trading partners. In contrast, this Figure is 22% in the extra-high-voltage grid.

A further effect of the structure on energy trading can be found in Figure 4.22. For both grids, this shows the proportion of agents who did not receive their desired offer in the respective trading step and had to seek other offers or cease trading.

In the EHV network, there are many conflicts over centralised cheap offers. Due to the meshed structure, many agents try to fall back on these regional offers. It culminates in trading step three, in which around 84% of all agents did not receive their desired offer because it had already been used up.

This stands in clear contrast to the chain-like structure of the MV network. Its structure means fewer conflicts, as an average of two neighbouring agents only claim a cheap central offer. Figure 4.22 also shows this, as in trading step two, only a maximum of 52% of agents do not receive their desired offer. Across all trading steps, agents in the MV network are more likely to receive their desired offer than in the EHV network.



Figure 4.22.: Proportion of agents missing a trade

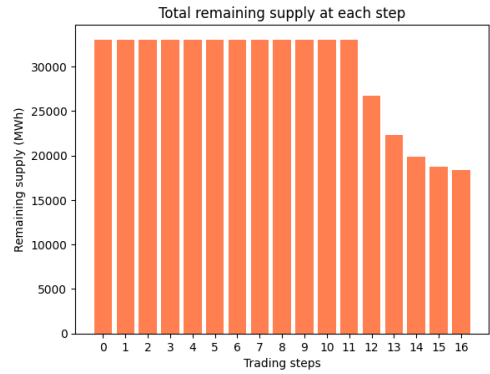


Figure 4.23.: Total remaining supply (MV)

All observations summarised, the structure of the network naturally also influences the successfully matched amount of generation. For example, the EHV model

matches half of the possible generation, while the MV model only consumes around 44% of the generation. Compare Figure 4.6 to Figure 4.23. The same applies identically to the loads due to the equal distribution of generation and consumption.

4.5. Summary

To summarise, the approach converges for both network structures. This is shown by the figures for matched supply in the EHV and MV grid and the Figure for the proportionate reaching of the local clearing price by the agents. As soon as an agent has reached its local clearing price, i.e., the merit order price of the cell, energy trading is automatically cancelled, no further energy is traded, and no further interaction with agents is performed in the same trading step. Therefore, the grid's structure only influences the time of convergence. A grid with high transport costs, i.e., a high impedance, converges faster than a grid with lower transport costs. Meshing also plays a central role. On the one hand, this means that more agents do not receive their desired trade and instead have to accept more expensive trades that are potentially further away. It leads to the merit order price being reached more quickly. On the other hand, it also invalidates this observation, as agents in a meshed network can initially see more trading partners. This means that many potentially favourable supplies can be requested.

Conversely, due to its structure, the opposite is true in the chain-like MV network. The network type initially provides fewer trading partners but also fewer conflicts over demanded supplies overall. Which of the two observations (impedance or meshing) has a more significant impact on the convergence of energy trading cannot be conclusively assessed. However, it can be observed that both play a central role in the speed at which convergence occurs.

Regionality is also maintained in both grids. With a few exceptions, the load is mainly balanced with supply from the own cell or direct neighbours. The individual exceptions also do not exceed the previously determined local trading cap. Once again, the structure plays a central role in determining the extent of trading. The local trading cap, determined in the Explore phase, depends on the structure. The actual trade, however, is very dependent on transport costs. In the EHV grid, more trade takes place with direct neighbours, which also influences the general coverage of all loads in the network. In the MV network, on the other hand, only local trade takes place. The MV grid is more regional than the EHV grid. However, this happens at the expense of the MV Grid's load coverage.

Therefore, although regionality must play a central role, too much enforced regional trading can also jeopardise load coverage.

5. Conclusion

This thesis demonstrated a possible approach to decentralised, self-organised control of the electricity grid. The VDE concept of a cellular energy system was expanded to include the principles of self-organisation.

In order to implement complete decentralisation, the decentralised regional consumption of generated energy was also taken into account. For this purpose, a corresponding energy trading system that prices grid losses was incorporated into the approach already mentioned.

It shows that such a concept can converge and trade its energy regionally. Furthermore, even a simple random distribution of the possible maximum costs for generated power and its associated transport costs can lead to load coverage of 50% in a meshed grid.

The modified merit order principle developed as part of the new type of energy trading also means that each cell and, therefore, sub-grids can calculate their dynamic grid price. It is in clear contrast to the current centralised and global market price. The advantage of this is that cells can now respond more specifically to their individual needs and can regulate and determine their generation and load coverage using their own price. In addition, the newly developed approach promotes the reduction of grid losses through the pricing of transport costs and also prevents the large-scale unwanted expansion of the grid that would otherwise be necessary.

A sensible distribution of the maximum price must now be found in further work. Considering realistic load and generation amounts would be a further logical step for a subsequent simulation. As this approach only deals with trading between cells of the same cell level, a further approach to be investigated is trading or communication between higher and lower-level cells. Investigating how the approach behaves in exceptional situations such as grid failures would also be interesting.

A. List of Abbreviations

TSO Transmission System Operators

DSO Distribution System Operators

VDE German Association for Electrical Engineering, Electronics and Information Technology

EHV Extra-high voltage grid

MV Medium voltage grid

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